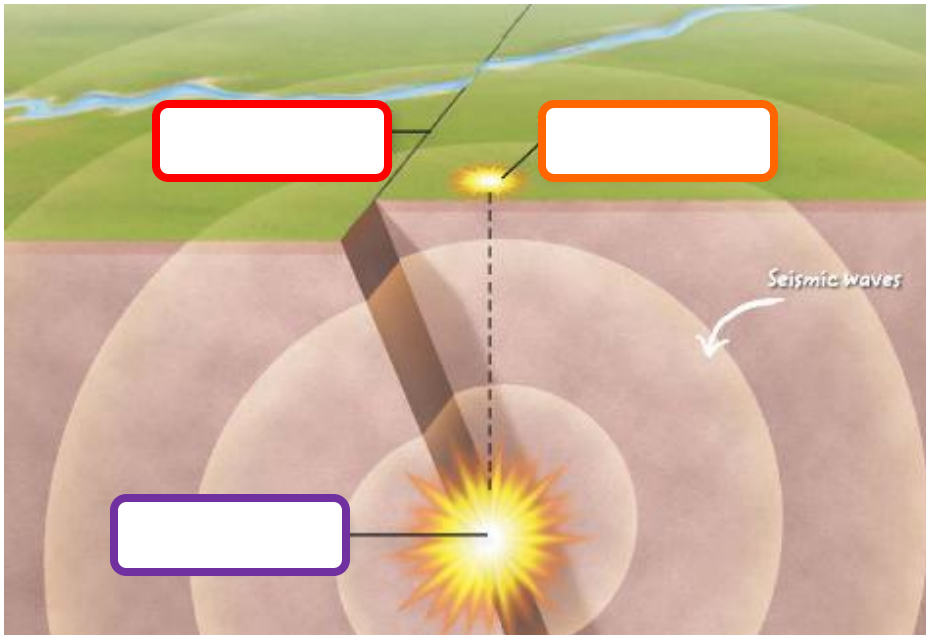


Earthquakes



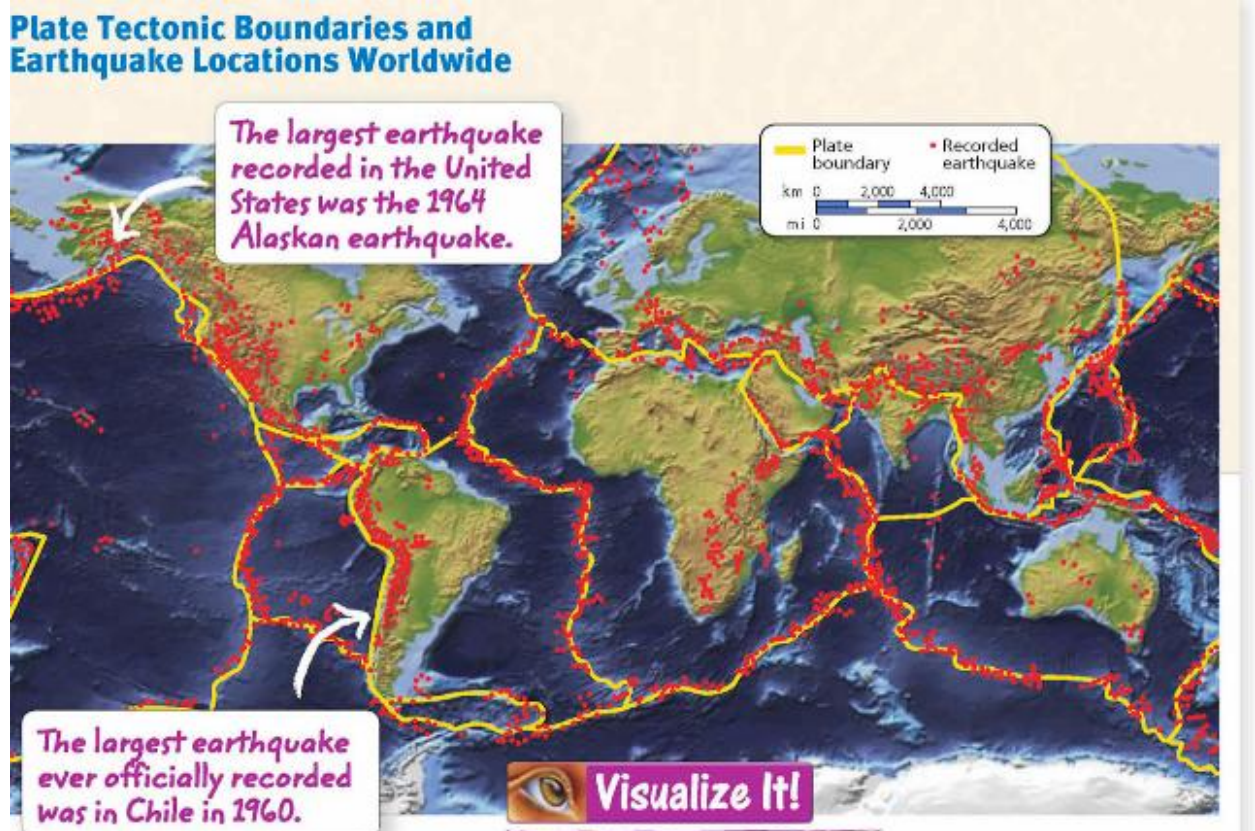
Label the following:

FAULT

EPICENTER

FOCUS

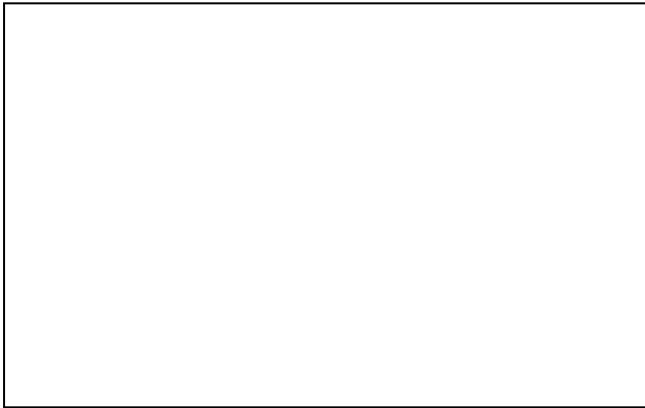
CIRCLE
NORTH
AMERICA



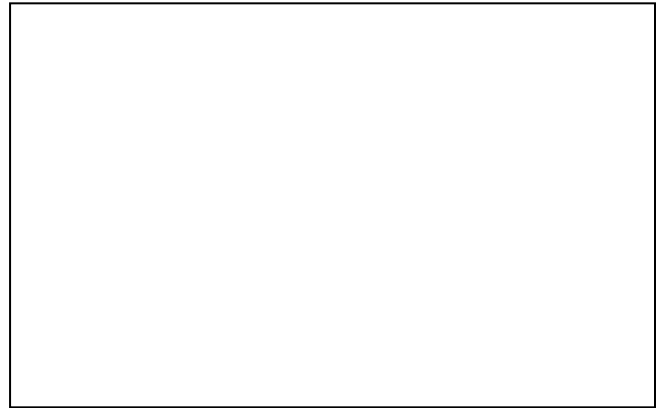
Most earthquakes are (**NEAR PLATE BOUNDARIES** / **FAR AWAY FROM PLATE BOUNDARIES**).

North America has **MANY** earthquakes. True or false? (**TRUE** / **FALSE**)

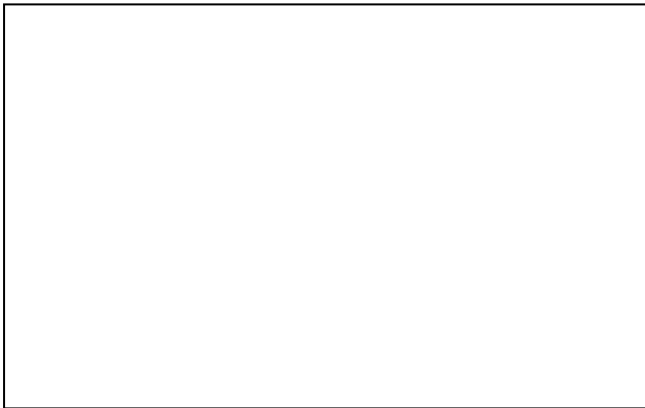
Cut and paste or Velcro the appropriate plate location terms with their corresponding image. Then, **Circle** in **RED** the earthquake epicenters:



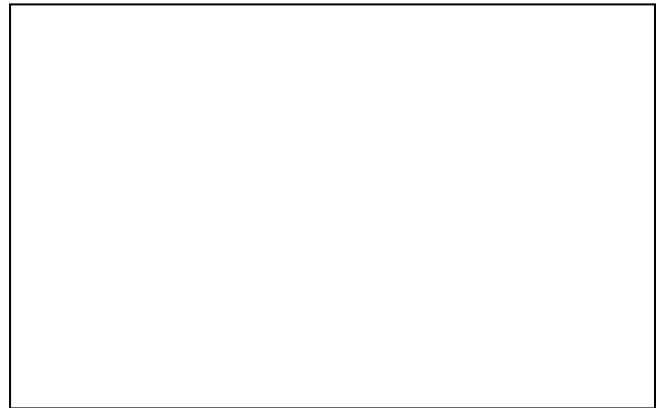
Divergent Boundary



Convergent Boundary



Transform Boundary

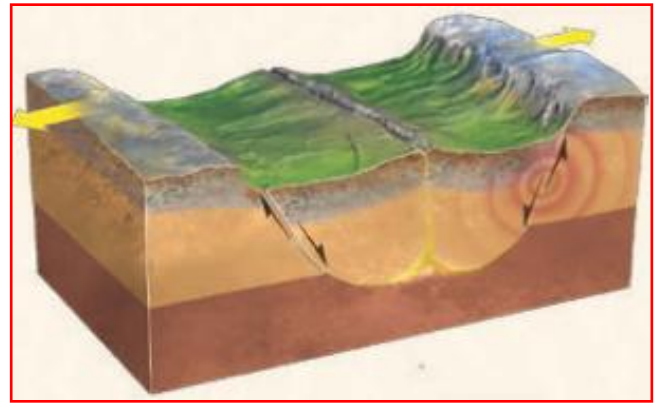


**Mid-Continental Zone (Old
Fault Zone)**

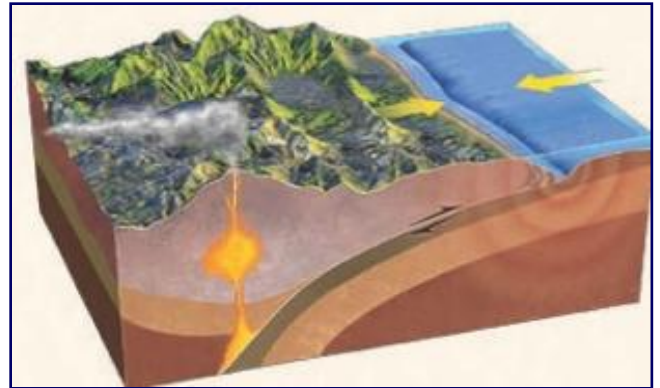
Most earthquakes occur where (**PLATES** / **OCEANS**) meet.

Sometimes earthquakes occur on old (**ICEBERGS** / **FAULTS**).

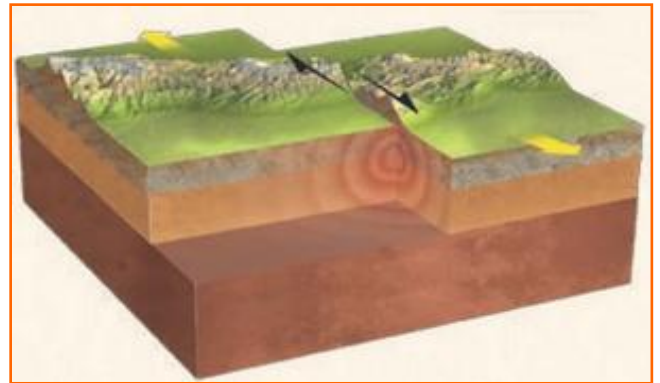
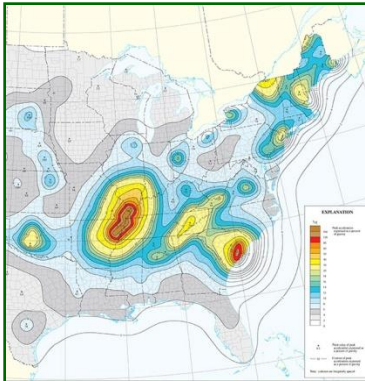
FAULT



EPICENTER



FOCUS



Locating an Earthquake Epicenter



Seismometer

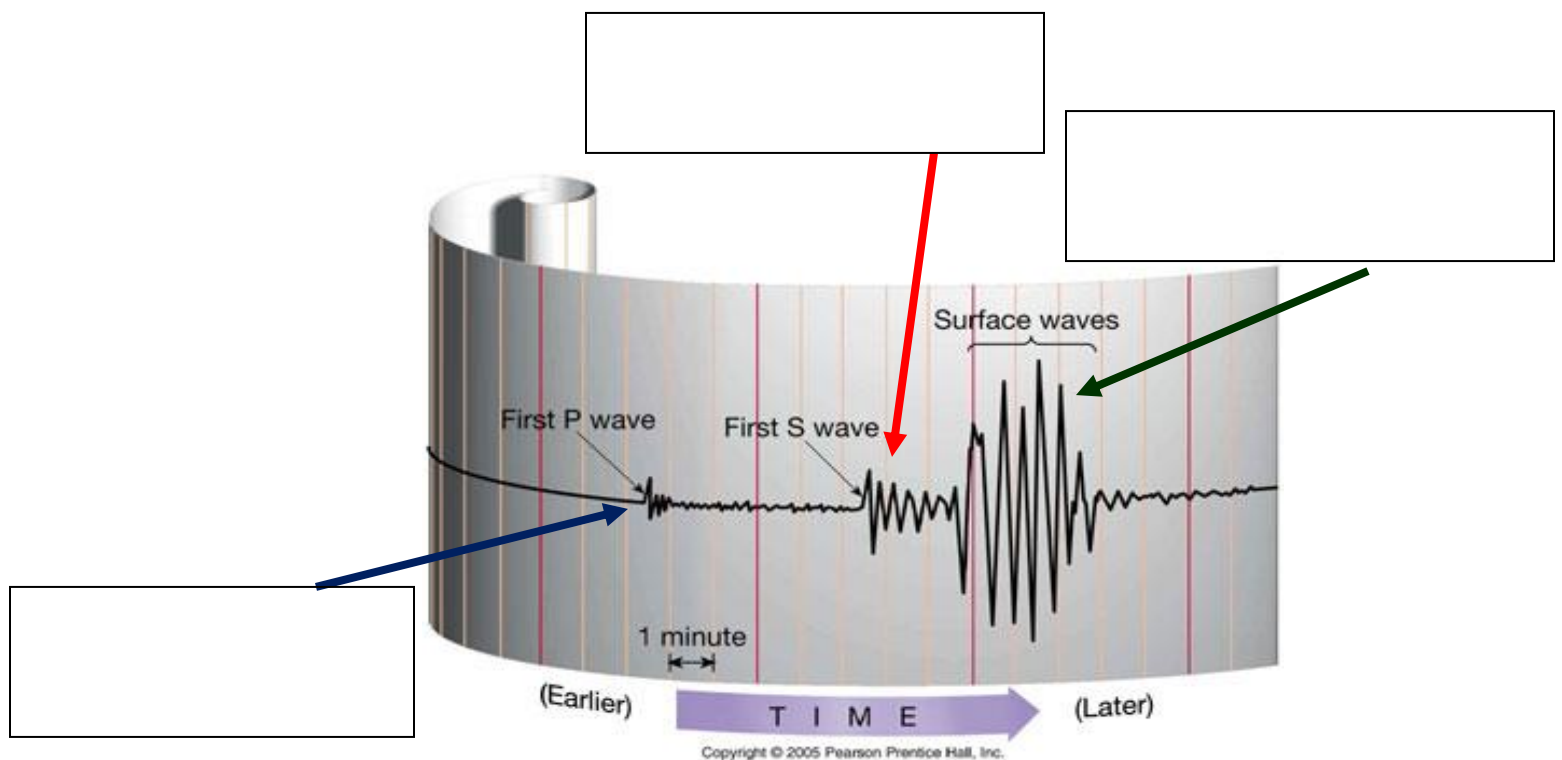


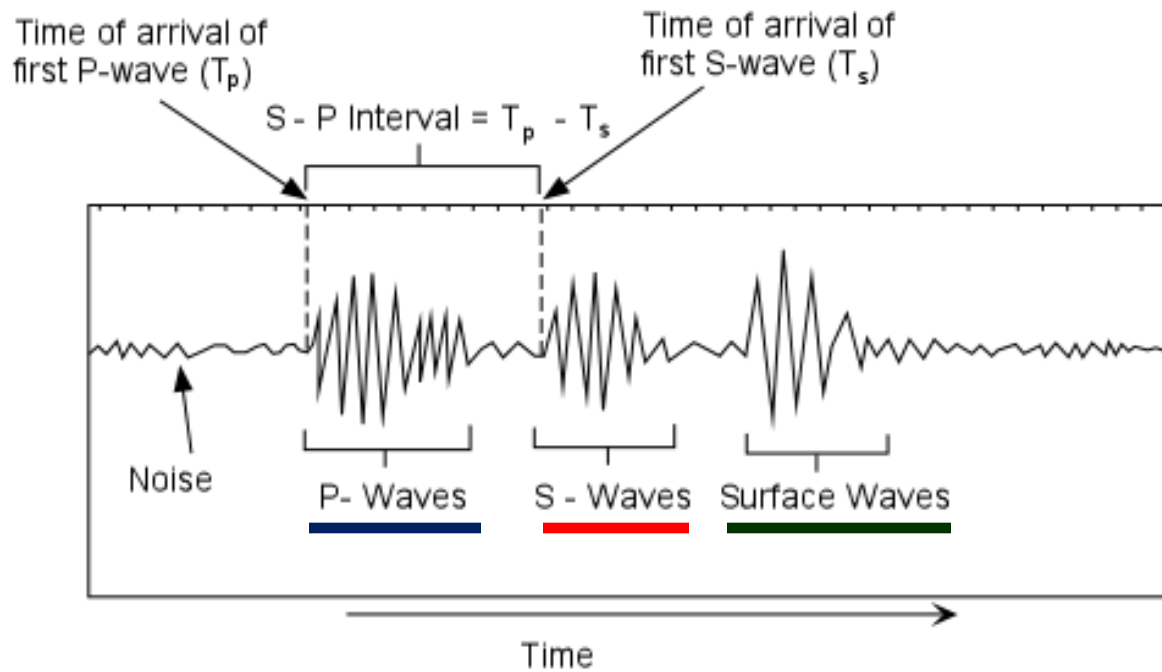
Seismograph

Circle in **BLUE** the P-wave.

Circle in **RED** the S-wave

Circle in **GREEN** the surface waves (L-waves)

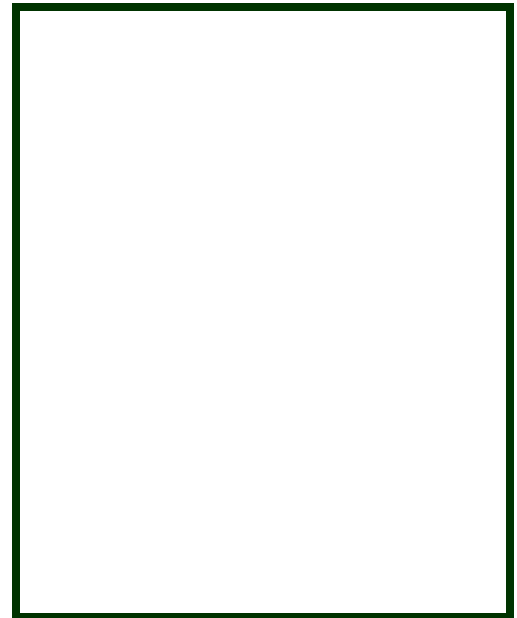
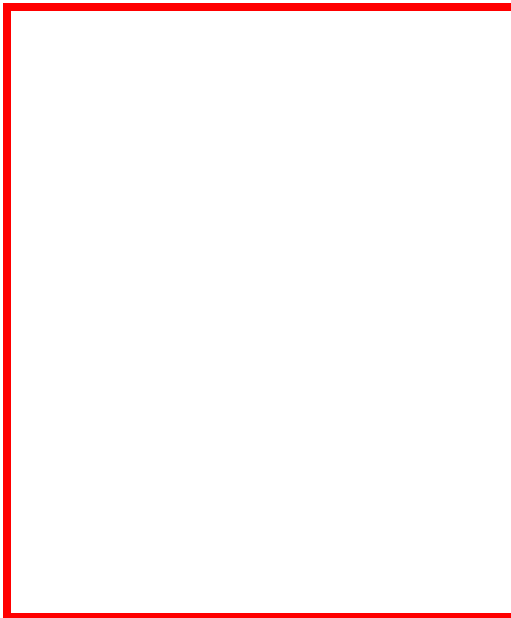
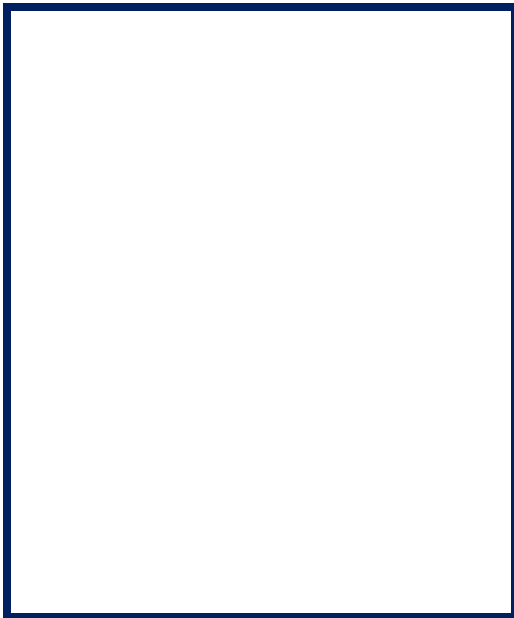




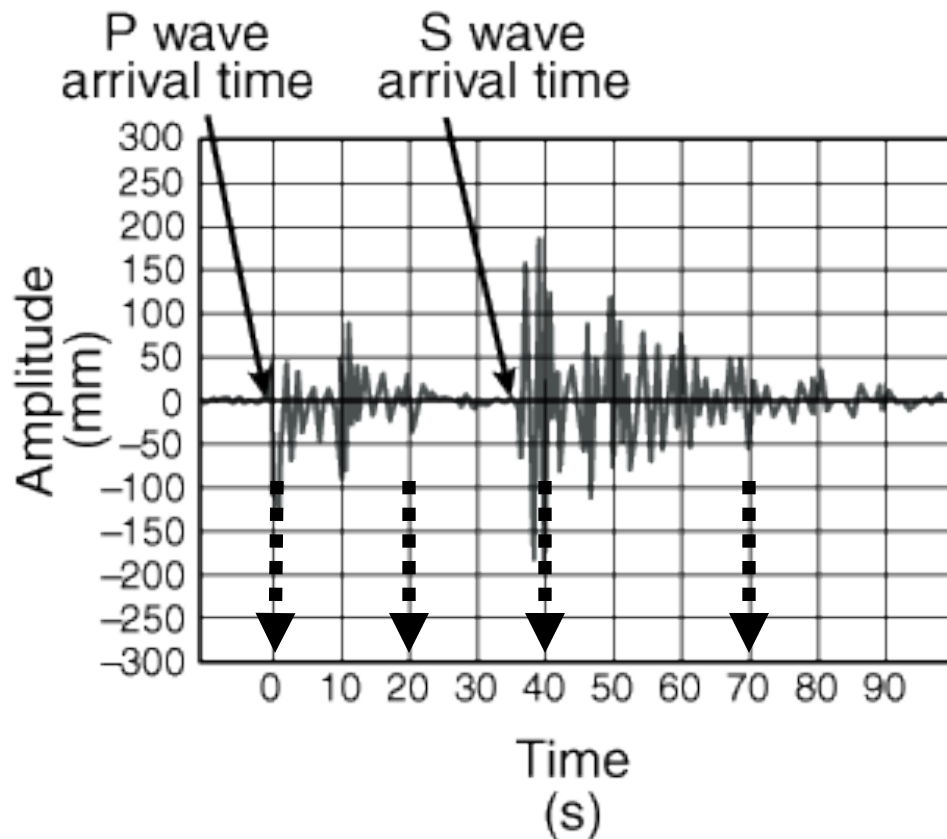
P-wave ("JOLT")

**S-wave ("UP and DOWN
and SIDE to SIDE")**

**LOVE wave ("SIDE to
SIDE")**



Seismogram



Trace the **P-wave** in **BLUE**

Trace the **S-wave** in **RED**



WRITE

or RUBBER STAMP



BELOW

The **P-wave** begins at time _____ seconds

The **P-wave** ends at time _____ seconds

The **S-wave** begins at time _____ seconds

The **S-wave** ends at time _____ seconds

Once again,

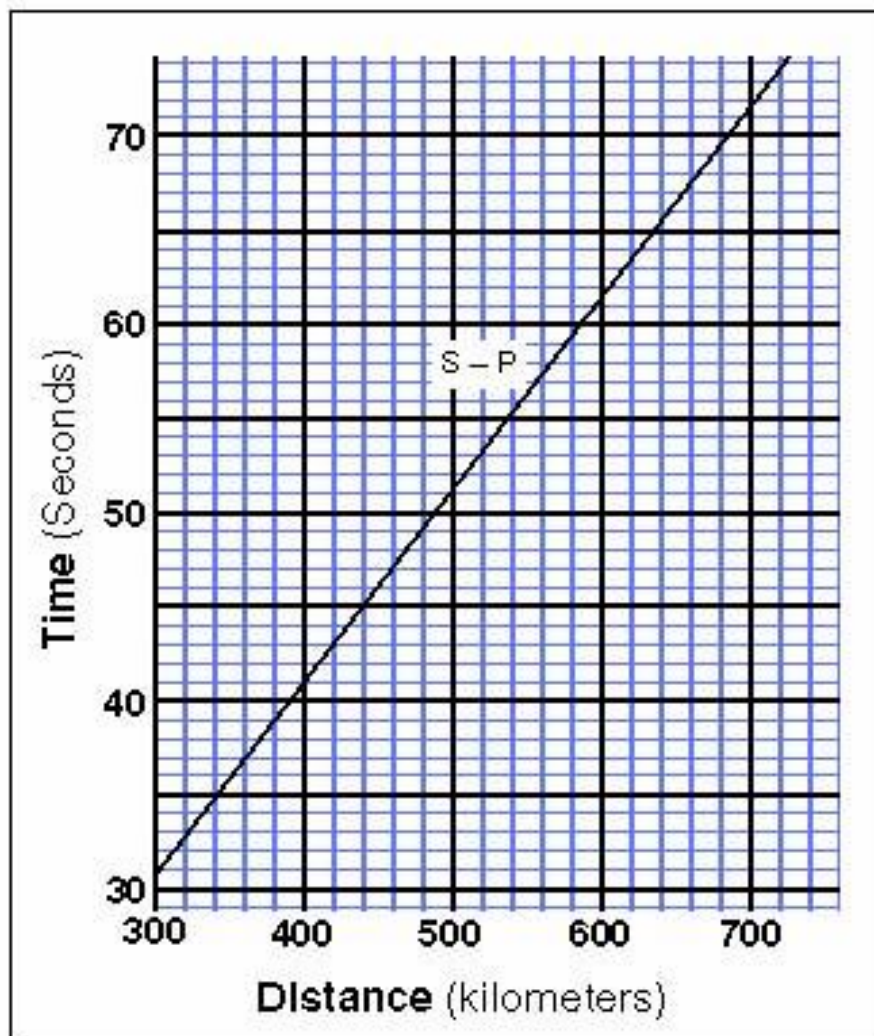
The **P-wave** began at time _____ seconds

The **S-wave** began at time _____ seconds

Now, subtract the **P-wave** time from the **S-wave** time:

S – P = Distance from Epicenter of Earthquake (km)

_____ - _____ = _____ km



Example Only

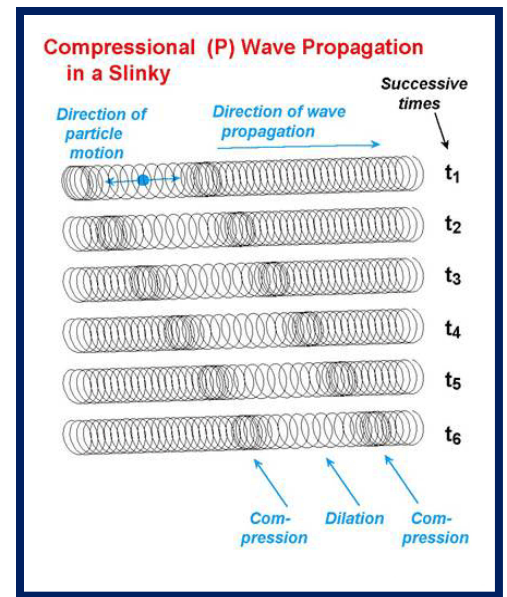
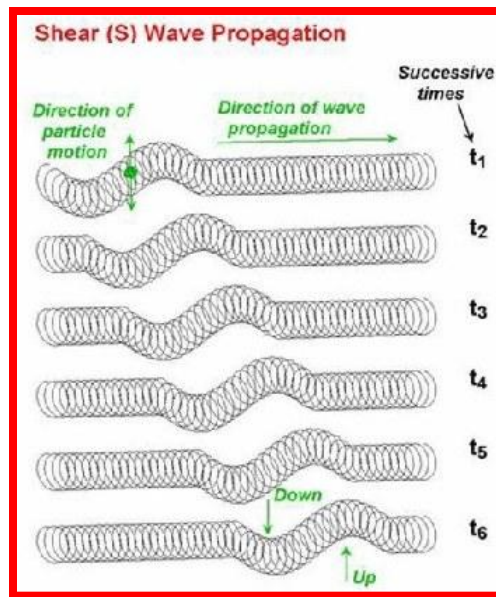
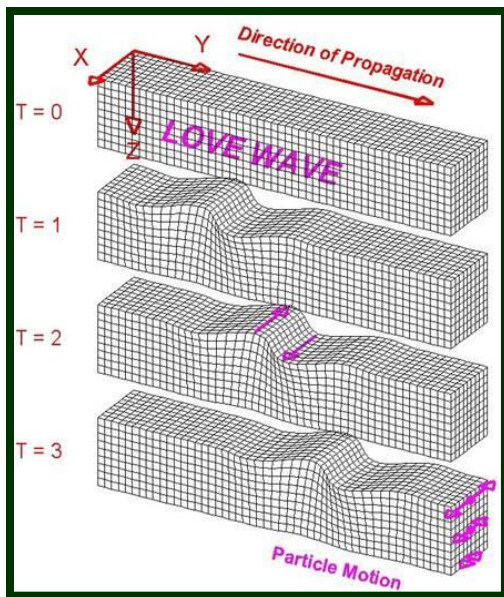


Trace the circles in their colors of **RED**, **BLUE**, and **GREEN**:



In which city do the 3 circles intersect?

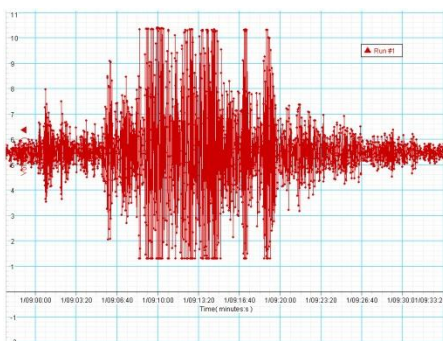
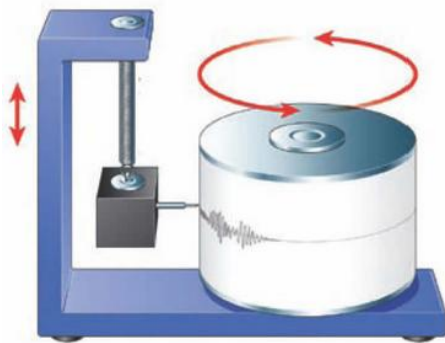
(**NEW YORK CITY** **OR** **LOS ANGELES**)



P-wave

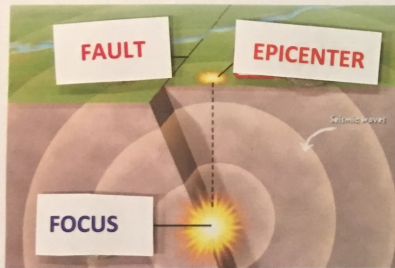
S-wave

Surface wave



Answer Key

Earthquakes



Label the following:

FAULT

EPICENTER

FOCUS

Plate Tectonic Boundaries and Earthquake Locations Worldwide



Most earthquakes are (**NEAR PLATE BOUNDARIES** / FAR AWAY FROM PLATE BOUNDARIES).

North America has **MANY** earthquakes. True or false? (**TRUE** / FALSE)

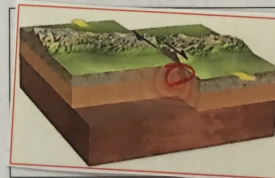
Cut and paste or Velcro the appropriate plate location terms with their corresponding image. Then, **Circle** in **RED** the earthquake epicenters:



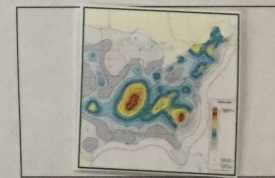
Divergent Boundary



Convergent Boundary



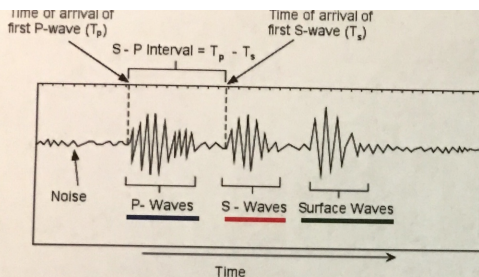
Transform Boundary



Mid-Continental Zone (Old Fault Zone)

Most earthquakes occur where (**PLATES** / OCEANS) meet.

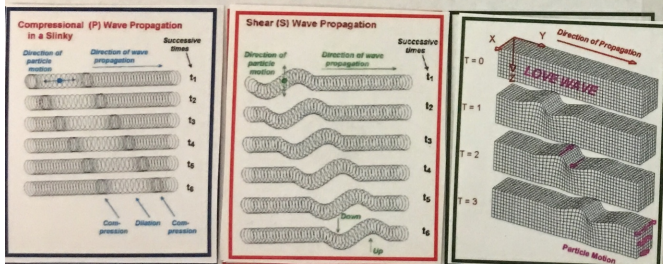
Sometimes earthquakes occur on old (**ICEBERGS** / **FAULTS**).



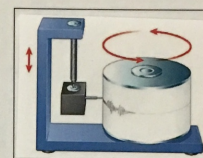
P-wave ("JOLT")

S-wave ("UP and DOWN and SIDE to SIDE")

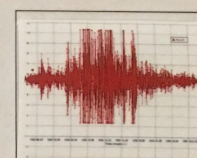
LOVE wave ("SIDE to SIDE")



Locating an Earthquake Epicenter



Seismometer



Seismograph

Circle in **BLUE** the P-wave.

Circle in **RED** the S-wave

Circle in **GREEN** the surface waves (L-waves)

